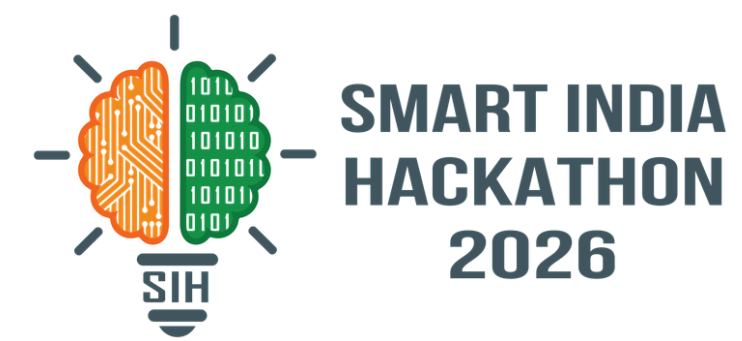
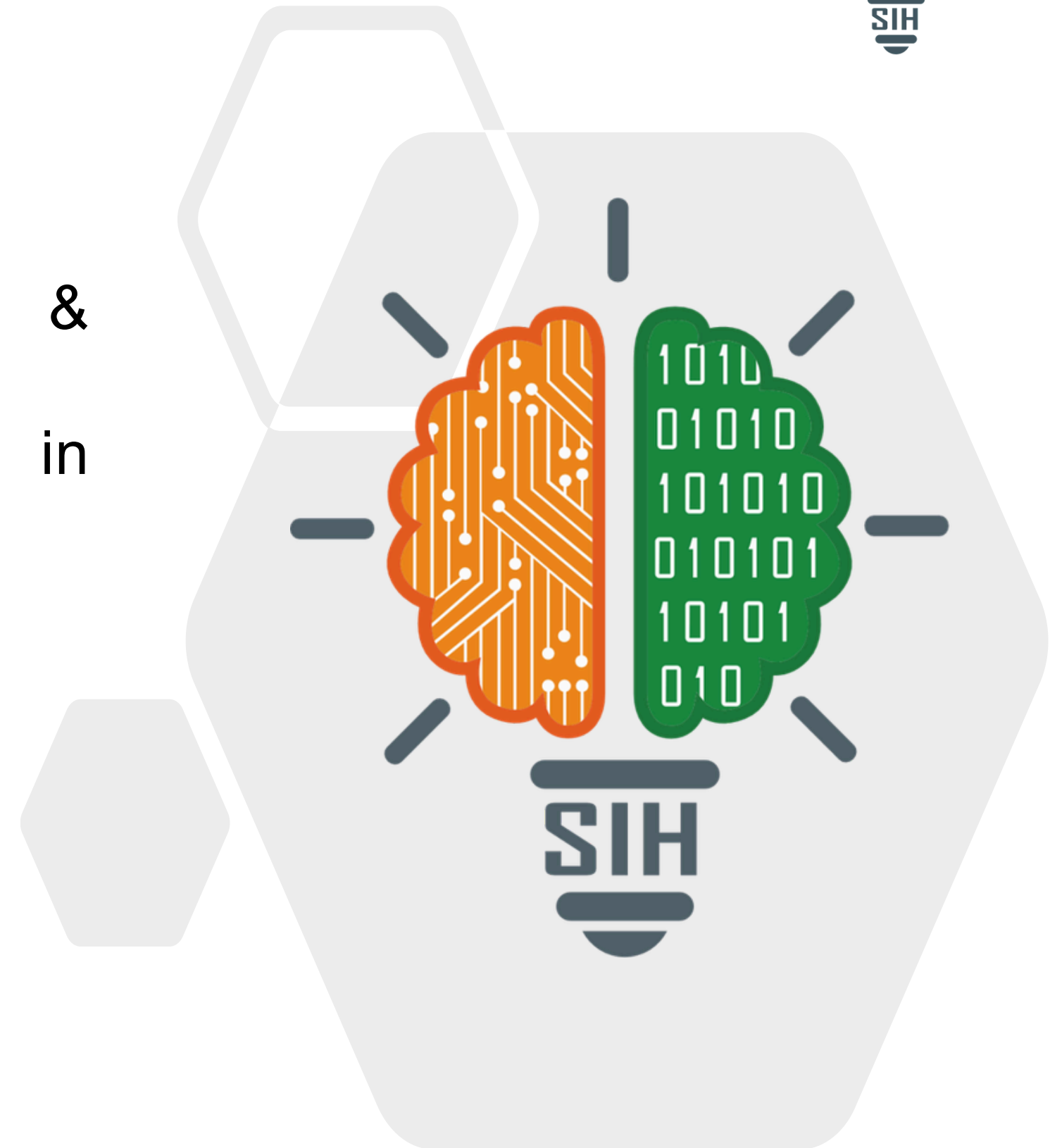


# SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26133
- **Problem Statement Title**- Accessibility & Quality of Public Healthcare in Rural/Underserved Areas
- **Theme**- MedTech / BioTech / HealthTech
- **PS Category**- Software
- **Team ID**- 129855
- **Team Name**- The Glitch Gang





## Our Vision

“ We are building a rural community healthcare platform that improves accessibility and quality of public healthcare through community health monitoring, early risk detection, and timely connection to healthcare services.”

## Key Features



### Community Health Reporting

Citizen + ASHA/ANM reporting, simple forms, offline support



### Water Quality Monitoring

pH, turbidity, TDS, temperature & more



### Environmental Data

Weather, external APIs, historical trends



### AI/ML Risk Prediction

Combine data, find patterns, get risk signals



### Alerts

App, SMS, ASHA/ANM, government & location-based



### Government Dashboard

Risk map, water quality, health trends, real-time overview

## Target Users



Citizen /  
Rural Resident



ASHA / ANM /  
Health Worker



Government  
Health Official



System  
Administrator

## Supporting Stakeholders



Water resource  
personnel



District/state  
authorities



NGOs



Sensor operators



Public health  
organizations



Other relevant  
government stakeholders

## Key Goals



Better Access  
to Healthcare



Healthier  
Communities



Early Risk  
Detection



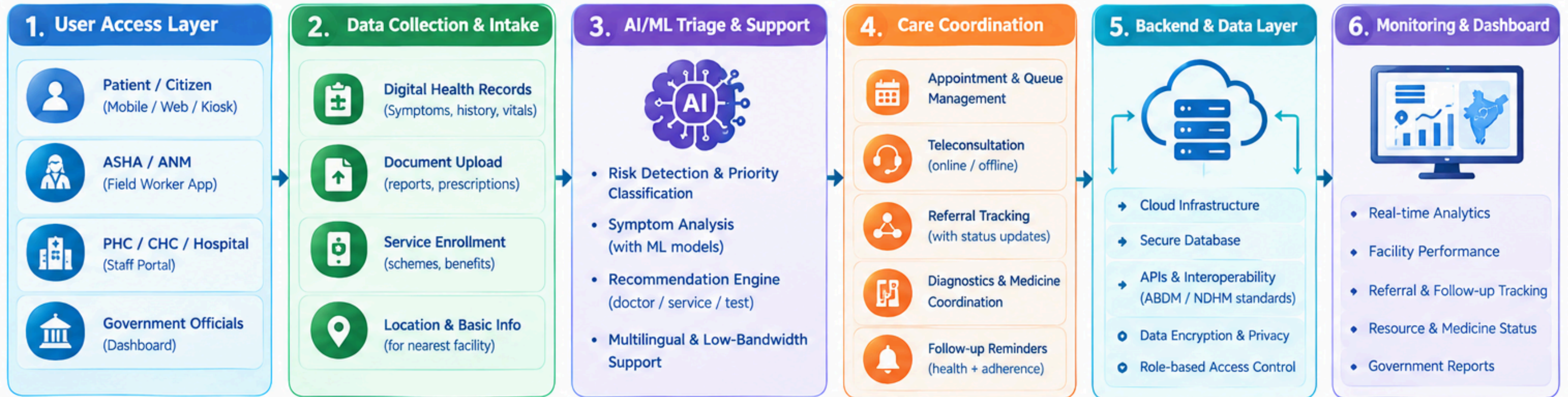
Stronger  
Rural Healthcare



Healthier People  
Stronger India

# TECHNICAL APPROACH

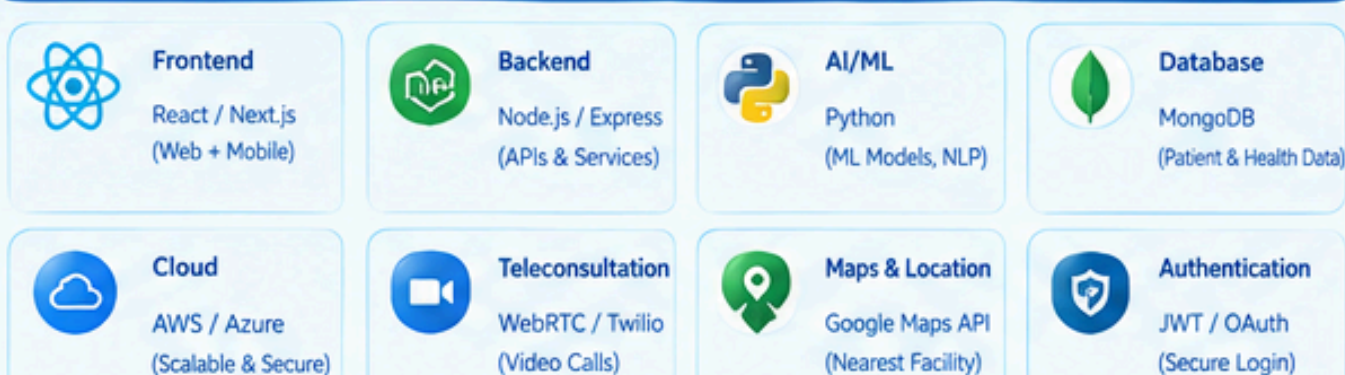
AI-Powered Rural Healthcare Access & Care Coordination Platform



## System Flow (Simplified)



## Key Technologies



## Integration with Existing Systems



## Key Technical Highlights

- ✓ AI-assisted triage (not a replacement for doctors)
- ✓ Works in low-connectivity environments
- ✓ Supports multiple languages (Marathi, Hindi + regional)
- ✓ Interoperable with approved health standards
- ✓ Secure, scalable and modular architecture
- ✓ Designed for rural and underserved regions

# FEASIBILITY AND VIABILITY

A Practical, Sustainable and Scalable Solution for Rural Public Healthcare



## 1. Technical Feasibility

Modern Tech | Secure | Scalable



### PWA / Web Application

- Responsive & device friendly
- Works on low-end devices



### Offline First

- Local storage + auto sync
- Works without internet



### Multilingual UI

- English + Hindi + Marathi + local languages
- Simple, easy-to-use interface



### Secure & Role-based Access

- Citizen / ASHA / ANM / Doctor / Admin
- Data privacy & security



### Reliable Tech Stack

- Supabase + PostgreSQL (data)
- FastAPI + Python (AI service)
- Modular & scalable architecture



## 2. Economic & Operational Feasibility

Cost Effective | Existing Infrastructure | Easy to Adopt



Reuse Existing PHC/CHC Facilities



Low Incremental Infrastructure Cost



Lightweight Workflows (ASHA/ANM/PHC)



Less Travel & Waiting Time



Phased Deployment (Starting Small)



## 3. Risk Assessment

Identify | Plan | Reduce



### Low Connectivity

→ Offline sync, local storage, low bandwidth mode



### Data Privacy & Security

→ RLS, audit logs, encryption



### AI Uncertainty

→ Human-in-the-loop, expert review



### Interoperability Integration

→ Standards-based (ABDM/NDHM), proper APIs



### Adoption & Training

→ Simple UI, local training, pilot testing



## 4. Innovation & Differentiation

Smarter Care | Better Coordination | Real Impact



### Care Coordination

Connects citizens, frontline workers, facilities and government



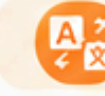
### AI-Assisted Triage / Priority Support

Helps identify priority cases (not a replacement for doctors)



### Closed-loop Referral Tracking

Follow-ups until completion



### Multilingual & Low-Connectivity Support

Works in rural conditions



### Longitudinal Health Records

Complete patient history across visits



Citizen / Patient  
(Registration & Intake)



ASHA / ANM  
(Home Visit & Support)



PHC / CHC  
(Primary / Community Health Centre)



Doctor / Teleconsultation  
(Consultation & Guidance)



### Dashboard



- ✓ Patients
- ✓ Referrals
- ✓ Appointments
- ✓ Diagnostics
- ✓ Medicines
- ✓ Follow-ups



Referral



Diagnostics /  
Medicines



Follow-up



Government  
Dashboard



## 5. Long-term Scalability

Bigger Reach | More Impact | Future Ready



More Facilities  
(PHC/CHC/District)



More Districts  
(Across Maharashtra)



More Languages  
(Local + Regional)



Full Interoperability  
(ABDM/NDHM Standards)



Expanded Teleconsultation  
(Experts & Specialists)



"Better Access • Better Coordination • Continuity of Care • Healthier Rural Communities"





### Target Users

People who need and support better healthcare access



#### Patients / Citizens

- Easy access to doctors & services
- Digital health records
- Awareness of schemes & benefits



#### ASHA / ANM (Community Health Worker)

- Simple and local language interface
- Support for home visits & follow-ups
- Quick data entry and tracking



#### Doctors / Specialists

- Teleconsultation support
- Access to patient history
- Better decision support



#### Government & Health Officials

- Real-time dashboards
- Monitor service delivery & outcomes
- Data for better planning and resource allocation



### Key Benefits

Direct impact on people and the healthcare system



#### Better Health Outcomes

Early detection and timely treatment  
reduce disease burden and mortality.



#### Clean and Safe Water

Improved hygiene and sanitation  
(prevents water-borne diseases).



#### Faster Response

Quick alerts, referrals and resource  
allocation for urgent cases.



#### Empowered Communities

Greater awareness, access to services  
and health literacy.



#### Data-Driven Governance

Transparent monitoring, better decisions  
and efficient use of resources.



### Overall Impact

A stronger, more accessible and  
accountable rural healthcare system.



Improved  
Access to  
Healthcare



Continuity  
of Care



Higher Referral  
Completion  
Rates



Better Medicine &  
Diagnostic Availability  
Visibility



Improved  
Quality  
Monitoring



Reduced Travel  
& Waiting Time

## Our Research

We studied real-world problems, government initiatives and existing solutions to understand the current healthcare ecosystem in rural and underserved areas.



### Government Reports & Policies

- National Health Mission (NHM)
- Ayushman Bharat
- Health & Wellness Centres (HWC)



### Research Papers & Academic Studies

- Rural healthcare access
- Digital health & telemedicine
- AI/ML in healthcare



### Existing Systems

- eSanjeevani (Teleconsultation)
- ABDM (Health ID)
- CoWIN (Digital health infrastructure)



### User Insights

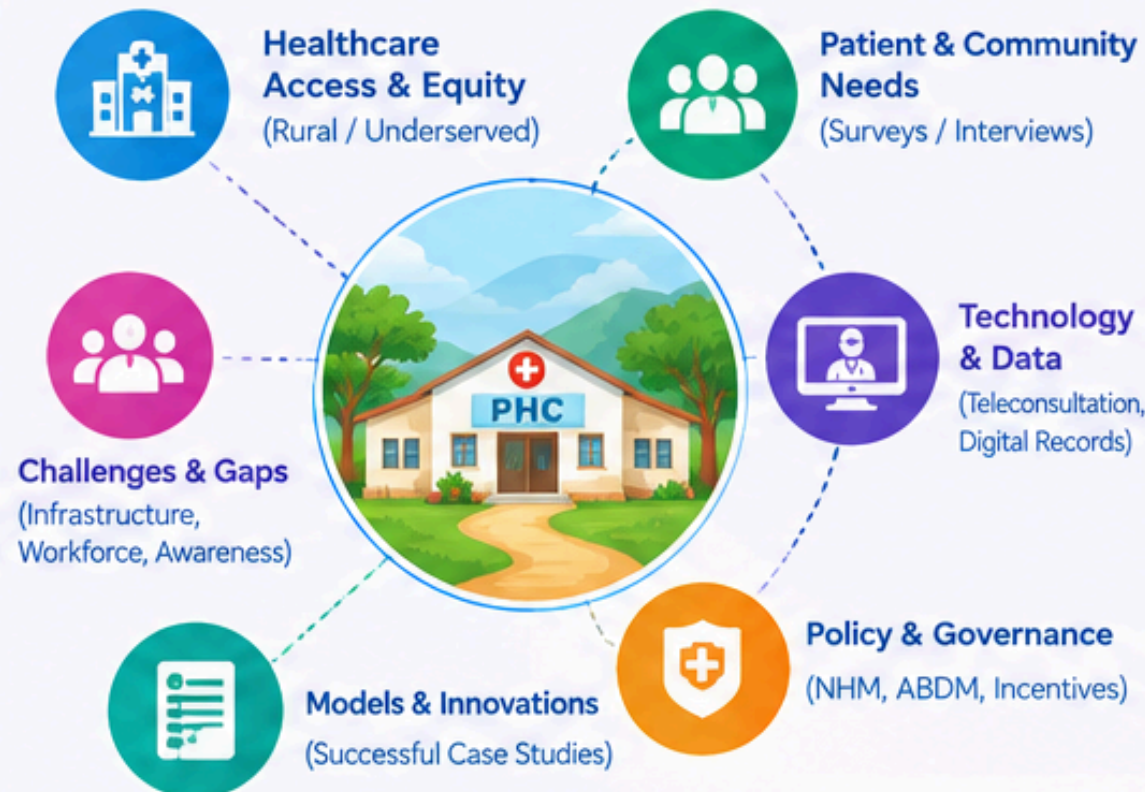
- Interviews with ASHA/ANM workers
- Feedback from rural residents
- Insights from healthcare professionals



### Best Practices

- Successful rural health models (e.g., NRHM)
- Digital health platforms
- Community health worker integration

## Our Research Focus



**Goal:** Understand the real needs, challenges and opportunities to create a practical, scalable and impactful solution for rural healthcare.



## References and Sources



**Government Portals**  
(NHM, ABDM, MoHFW)



**Research Papers**  
(IEEE, Springer, etc.)



**Web Resources**  
(WHO, UNICEF, NITI Aayog, etc.)



**Field Interviews**  
(ASHA / ANM / Citizens)



**Existing Platforms**  
(eSanjeevani, CoWIN, ABDM)

## Key Findings & Takeaways

- ✓ **Access is limited** – Many rural areas still lack adequate healthcare infrastructure and specialists.
- ✓ **Digital solutions help** – Teleconsultation, health apps and digital records can reduce distance and time barriers.
- ✓ **Community health workers are vital** – ASHA/ANM play a key role in early detection and awareness.
- ✓ **Data and technology can improve decisions** – AI/ML and real-time data help in better triage, resource allocation and follow-up.
- ✓ **Awareness and trust matter** – People need simple, local-language and easy-to-use solutions.

## Why This Matters

- ♥ **Saves Lives**  
Early detection and timely treatment reduce disease burden and mortality.
- ♥ **Empowers Communities**  
Better access to healthcare builds healthier, stronger and more independent communities.
- ♥ **Builds a Healthier Future**  
A connected and inclusive healthcare system leads to long-term social and economic growth.

